CNNP SKILL-2 LAB EXPERIMENT

THREE-POINT CAPTURE – Analyze packet captures from three points in a network path to understand packet flow and identify performance differences.

Stage 1 – Install the required tools

```
125 packages can be upgraded. Run 'apt list --upgradable' to see them.
buntu@KOUSHAL:~$ sudo apt install -y tshark iperf3 mtr-tiny iproute2 w
ireshark netcat-openbsd
netcat-openbsd is already the newest version (1.234-1).
netcat-openbsd set to manually installed.
Upgrading:
  iproute2  libpcre2-8-0  libprotobuf-lite32t64  zlib1g

Installing:
  iperf3  mtr-tiny  tshark  wireshark

Installing dependencies:
  gnome-themes-extra-data  libqt6quick6
  ibverbs-providers        libqt6svg6
  libaacs0                 libqt6waylandclient6
  libabsl20260107          libqt6waylandcompositor6
  libavcodec62             libqt6widgets6
  libavformat62            libqt6wlshellintegration6
  libavutil60              librabbitmq4
  libb2-1                  librav1e0.8
  libbcg729-0              libre2-11
  libbdplus0               librist4
  libbluray3               libsbc1
  libchromaprint1          libsctp1
  libcjson1                libshine3
  libcodec2-1.2            libsm6
  libdav1d7                libsmi2t64
  libdouble-conversion3    libsnappy1v5
  libdvdnav4               libsoxr0
  libdvdread8t64           libspandsp2t64
```

```
Setting up ibverbs-providers:amd64 (61.0-2ubuntu3) ...
Setting up iperf3 (3.20-2) ...
Setting up libgrpc++1.51t64:amd64 (1.51.1-8ubuntu1) ...
Setting up libtbb12:amd64 (2022.3.0-2) ...
Setting up libqt6gui6:amd64 (6.10.2+dfsg-7) ...
Setting up qt6-gtk-platformtheme:amd64 (6.10.2+dfsg-7) ...
Setting up libavcodec62:amd64 (7:8.0.1-3ubuntu2) ...
Setting up libfalcosecurity0t64:amd64 (0.20.0-4ubuntu0.1) ...
Setting up libqt6network6:amd64 (6.10.2+dfsg-7) ...
Setting up libpcap0.8t64:amd64 (1.10.6-1ubuntu1.1) ...
Setting up qt6-qpa-plugins:amd64 (6.10.2+dfsg-7) ...
Setting up libavformat62:amd64 (7:8.0.1-3ubuntu2) ...
Setting up libqt6qml6:amd64 (6.10.2+dfsg-3) ...
Setting up libqt6opengl6:amd64 (6.10.2+dfsg-7) ...
Setting up libqt6qmlmodels6:amd64 (6.10.2+dfsg-3) ...
Setting up libqt6widgets6:amd64 (6.10.2+dfsg-7) ...
Setting up libqt6qmlworkerscript6:amd64 (6.10.2+dfsg-3) ...
Setting up libqt6svg6:amd64 (6.10.2-2) ...
Setting up libwireshark19:amd64 (4.6.4-1) ...
Setting up libqt6waylandclient6:amd64 (6.10.2+dfsg-7) ...
Setting up qt6-svg-plugins:amd64 (6.10.2-2) ...
Setting up libqt6qmlmeta6:amd64 (6.10.2+dfsg-3) ...
Setting up libqt6wlshellintegration6:amd64 (6.10.2+dfsg-7) ...
Setting up wireshark-common (4.6.4-1) ...
Setting up libqt6printsupport6:amd64 (6.10.2+dfsg-7) ...
Setting up tshark (4.6.4-1) ...
Setting up libqt6quick6:amd64 (6.10.2+dfsg-3) ...
Setting up libqt6multimedia6:amd64 (6.10.2-2) ...
Setting up wireshark (4.6.4-1) ...
Setting up libqt6waylandcompositor6:amd64 (6.10.2-4) ...
Setting up qt6-wayland:amd64 (6.10.2-4) ...
Processing triggers for libc-bin (2.43-2ubuntu2.4) ...
Processing triggers for man-db (2.13.1-1build1) ...
Processing triggers for libselinux1:amd64 (3.9-4build1) ...
Processing triggers for shared-mime-info (2.4-5build3) ...
Processing triggers for udev (259.5-0ubuntu3.4) ...
Processing triggers for hicolor-icon-theme (0.18-2build1) ...
buntu@KOUSHAL:~$
```

Stage 2 – Create the network

```
buntu@KOUSHAL:~$ sudo ip netns add host1
buntu@KOUSHAL:~$ sudo ip netns add router
buntu@KOUSHAL:~$ sudo ip netns add host2
buntu@KOUSHAL:~$ sudo ip link add veth-h1 type veth peer name veth-r1
buntu@KOUSHAL:~$ sudo ip link add veth-r2 type veth peer name veth-h2
buntu@KOUSHAL:~$ sudo ip link set veth-h1 netns host1
buntu@KOUSHAL:~$ sudo ip link set veth-r1 netns router
buntu@KOUSHAL:~$ sudo ip link set veth-r2 netns router
buntu@KOUSHAL:~$ sudo ip link set veth-h2 netns host2
buntu@KOUSHAL:~$
```

Stage 3 – Configure IP addresses

```
buntu@KOUSHAL:~$ sudo ip netns add host1
buntu@KOUSHAL:~$ sudo ip netns add router
buntu@KOUSHAL:~$ sudo ip netns add host2
buntu@KOUSHAL:~$ sudo ip link add veth-h1 type veth peer name veth-r1
buntu@KOUSHAL:~$ sudo ip link add veth-r2 type veth peer name veth-h2
buntu@KOUSHAL:~$ sudo ip link set veth-h1 netns host1
buntu@KOUSHAL:~$ sudo ip link set veth-r1 netns router
buntu@KOUSHAL:~$ sudo ip link set veth-r2 netns router
buntu@KOUSHAL:~$ sudo ip link set veth-h2 netns host2
buntu@KOUSHAL:~$ sudo ip netns exec host1 ip addr add 10.0.1.1/30 dev
veth-h1
buntu@KOUSHAL:~$ sudo ip netns exec host1 ip link set veth-h1 up


buntu@KOUSHAL:~$ sudo ip netns exec host1 ip link set lo up
buntu@KOUSHAL:~$ sudo ip netns exec router ip addr add 10.0.1.2/30 dev
 veth-r1
buntu@KOUSHAL:~$ sudo ip netns exec router ip addr add 10.0.2.1/30 dev
 veth-r2
buntu@KOUSHAL:~$ sudo ip netns exec router ip link set veth-r1 up
buntu@KOUSHAL:~$ sudo ip netns exec router ip link set veth-r2 up
buntu@KOUSHAL:~$ sudo ip netns exec router ip link set lo up
buntu@KOUSHAL:~$ sudo ip netns exec host2 ip addr add 10.0.2.2/30 dev
veth-h2
buntu@KOUSHAL:~$ sudo ip netns exec host2 ip link set veth-h2 up
buntu@KOUSHAL:~$ sudo ip netns exec host2 ip link set lo up
buntu@KOUSHAL:~$
```

Stage 4 – Enable routing

```
buntu@KOUSHAL:~$ sudo ip netns exec router sysctl -w net.ipv4.ip_forwa
rd=1
net.ipv4.ip_forward = 1
buntu@KOUSHAL:~$ sudo ip netns exec host1 ip route add 10.0.2.0/30 via
 10.0.1.2
buntu@KOUSHAL:~$ sudo ip netns exec host2 ip route add 10.0.1.0/30 via
 10.0.2.1
buntu@KOUSHAL:~$
```

Stage 5 – Test connectivity

```
buntu@KOUSHAL:~$ sudo ip netns exec host1 ping -c 3 10.0.2.2
PING 10.0.2.2 (10.0.2.2) 56(84) bytes of data.
64 bytes from 10.0.2.2: icmp_seq=1 ttl=63 time=0.129 ms
64 bytes from 10.0.2.2: icmp_seq=2 ttl=63 time=0.121 ms
64 bytes from 10.0.2.2: icmp_seq=3 ttl=63 time=0.068 ms

--- 10.0.2.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2045ms
rtt min/avg/max/mdev = 0.068/0.106/0.129/0.027 ms
buntu@KOUSHAL:~$
```

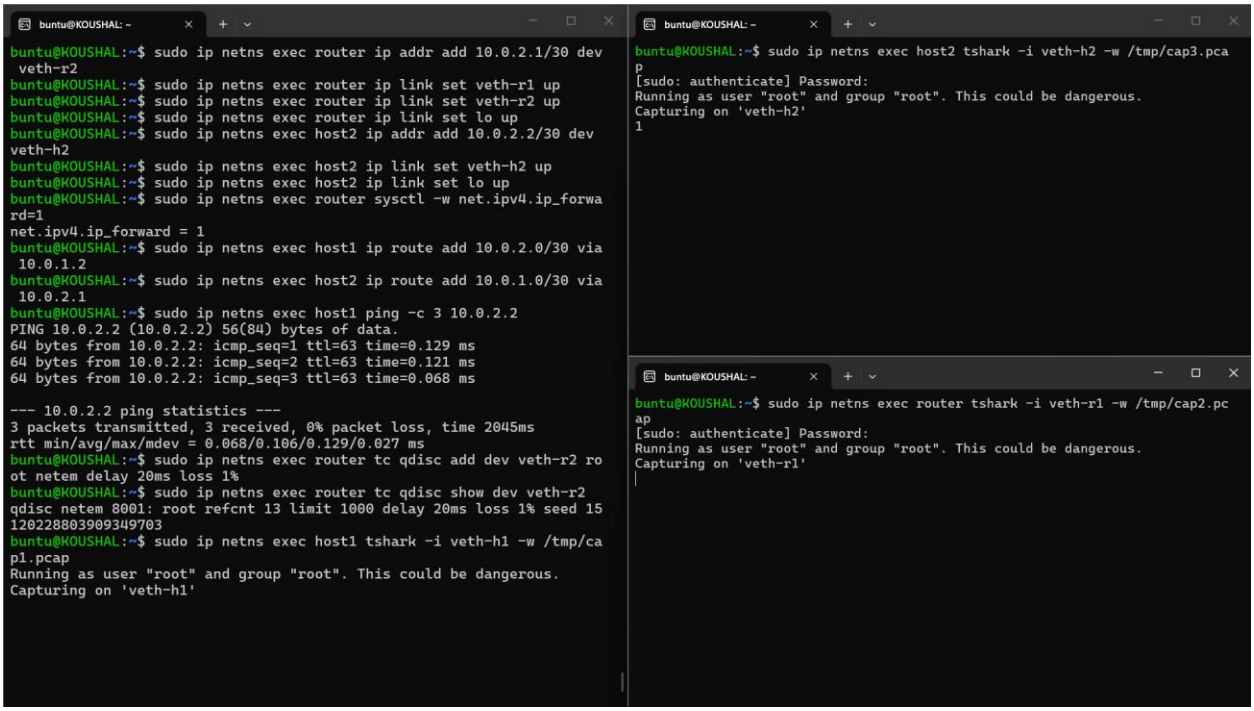
Stage 6 – Add network delay and packet loss

```
buntu@KOUSHAL:~$ sudo ip netns exec router tc qdisc add dev veth-r2 ro
ot netem delay 20ms loss 1%
buntu@KOUSHAL:~$ sudo ip netns exec router tc qdisc show dev veth-r2
qdisc netem 8001: root refcnt 13 limit 1000 delay 20ms loss 1% seed 15
120228803909349703
buntu@KOUSHAL:~$
```

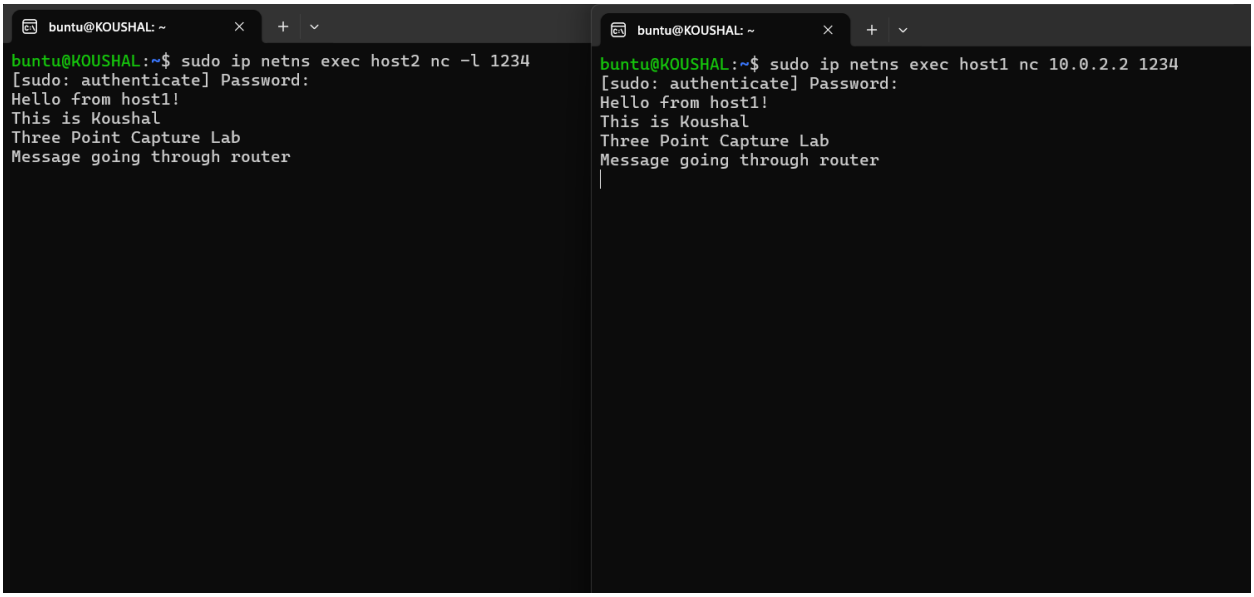
Stage 7 – Open five terminals

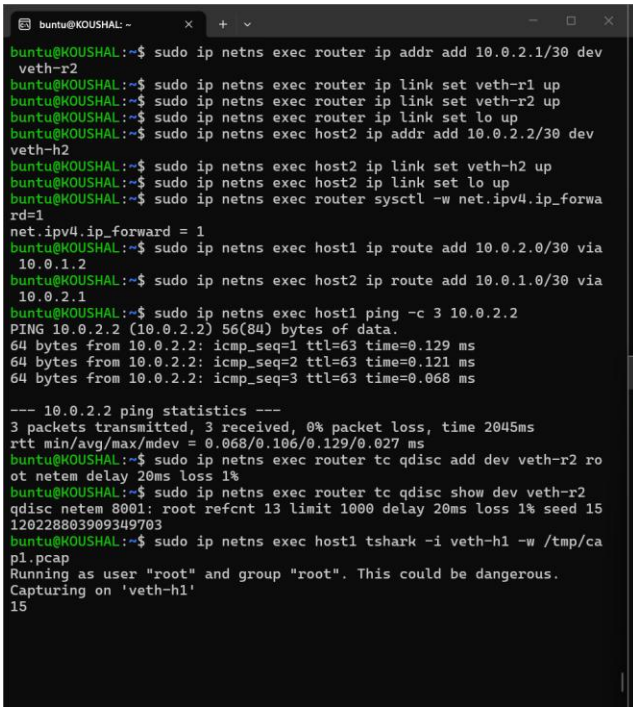
Create 4 new ubuntu terminals including the host terminal.

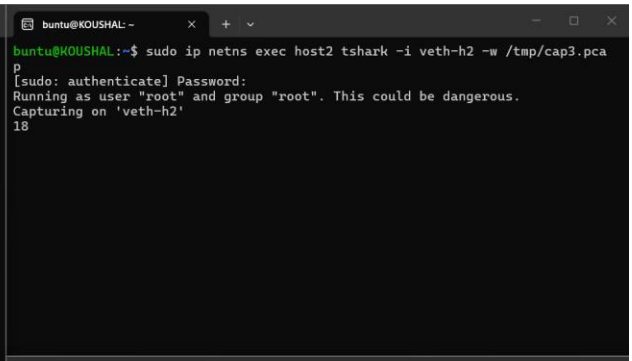
Stage 8 – Start three packet captures

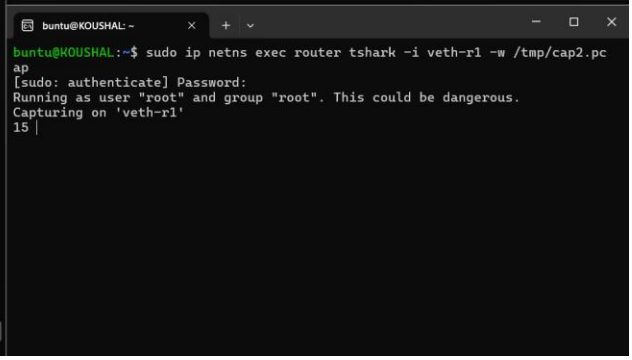


Stage 9 – Send a message from Host1 to Host2









Stage 10 – Stop packet capturing

```
Capturing on 'veth-h1'
17 ^C
buntu@KOUSHAL:~$ sudo chmod 644 /tmp/cap1.pcap /tmp/cap2.pcap /tmp/cap
3.pcap
buntu@KOUSHAL:~$ ls -lh /tmp/cap1.pcap /tmp/cap2.pcap /tmp/cap3.pcap
-rw-r--r-- 1 root root 2.1K Nov 12 05:16 /tmp/cap1.pcap
-rw-r--r-- 1 root root 2.1K Nov 12 05:16 /tmp/cap2.pcap
-rw-r--r-- 1 root root 2.2K Nov 12 05:16 /tmp/cap3.pcap
buntu@KOUSHAL:~$
```

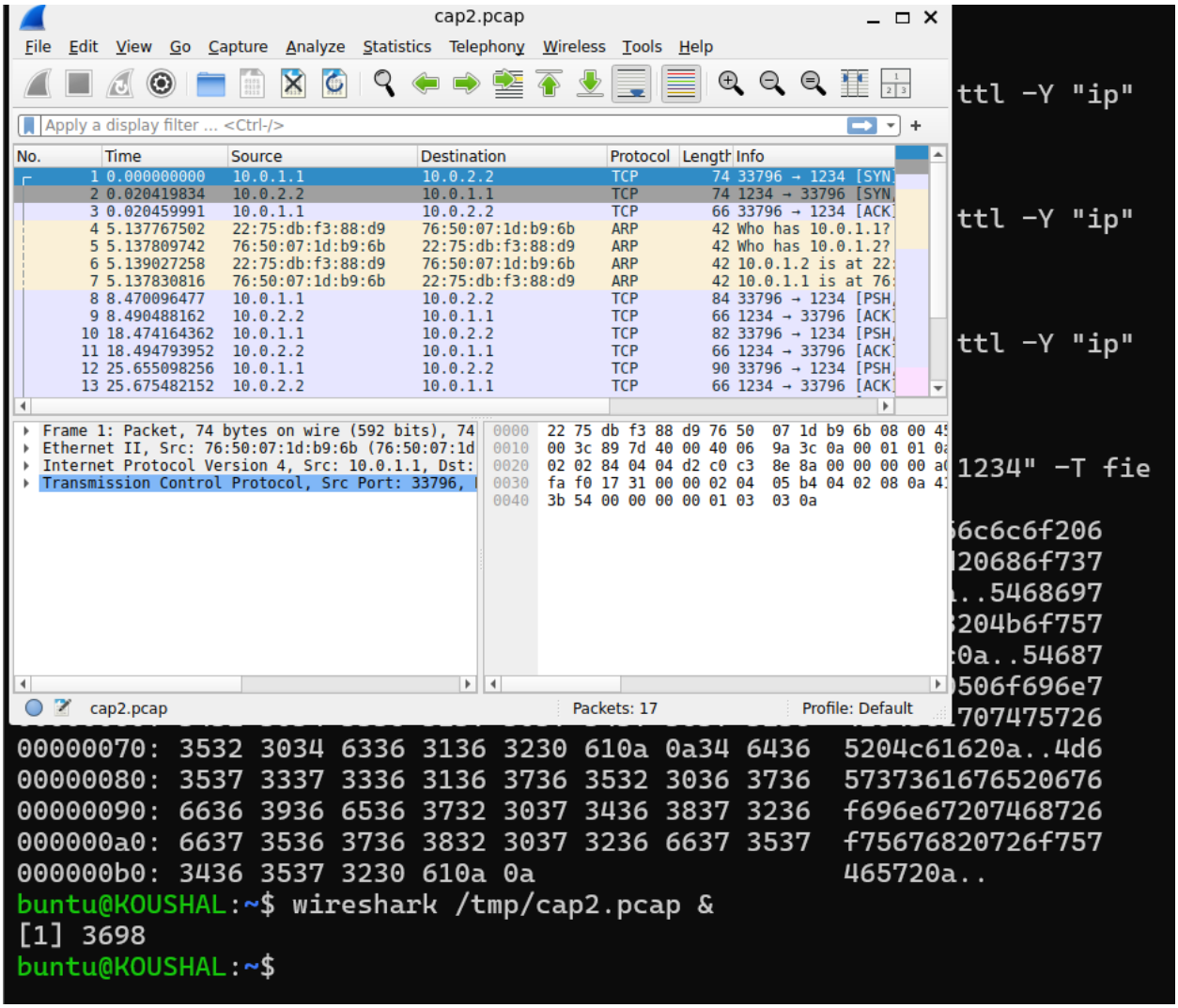
Stage 11 – Check TTL values

```
buntu@KOUSHAL:~$ tshark -r /tmp/cap1.pcap -T fields -e ip.ttl -Y "ip"
2>/dev/null | sort | uniq -c
      5 63
      6 64
buntu@KOUSHAL:~$ tshark -r /tmp/cap2.pcap -T fields -e ip.ttl -Y "ip"
2>/dev/null | sort | uniq -c
      5 63
      6 64
buntu@KOUSHAL:~$ tshark -r /tmp/cap3.pcap -T fields -e ip.ttl -Y "ip"
2>/dev/null | sort | uniq -c
      6 63
      5 64
buntu@KOUSHAL:~$
```

Stage 12 – Extract message data

```
buntu@KOUSHAL:~$ tshark -r /tmp/cap2.pcap -Y "tcp.port == 1234" -T fie
lds -e tcp.payload 2>/dev/null | xxd | head -20
00000000: 0a0a 0a34 3836 3536 6336 6336 6632 3036  ...48656c6c6f206
00000010: 3637 3236 6636 6432 3036 3836 6637 3337  6726f6d20686f737
00000020: 3433 3132 3130 610a 0a35 3436 3836 3937  431210a..5468697
00000030: 3332 3036 3937 3332 3034 6236 6637 3537  3206973204b6f757
00000040: 3336 3836 3136 6330 610a 0a35 3436 3837  368616c0a..54687
00000050: 3236 3536 3532 3035 3036 6636 3936 6537  2656520506f696e7
00000060: 3432 3034 3336 3137 3037 3437 3537 3236  4204361707475726
00000070: 3532 3034 6336 3136 3230 610a 0a34 6436  5204c61620a..4d6
00000080: 3537 3337 3336 3136 3736 3532 3036 3736  5737361676520676
00000090: 6636 3936 6536 3732 3037 3436 3837 3236  f696e67207468726
000000a0: 6637 3536 3736 3832 3037 3236 6637 3537  f75676820726f757
000000b0: 3436 3537 3230 610a 0a                   465720a..
buntu@KOUSHAL:~$
```

Stage 13 – Open Wireshark



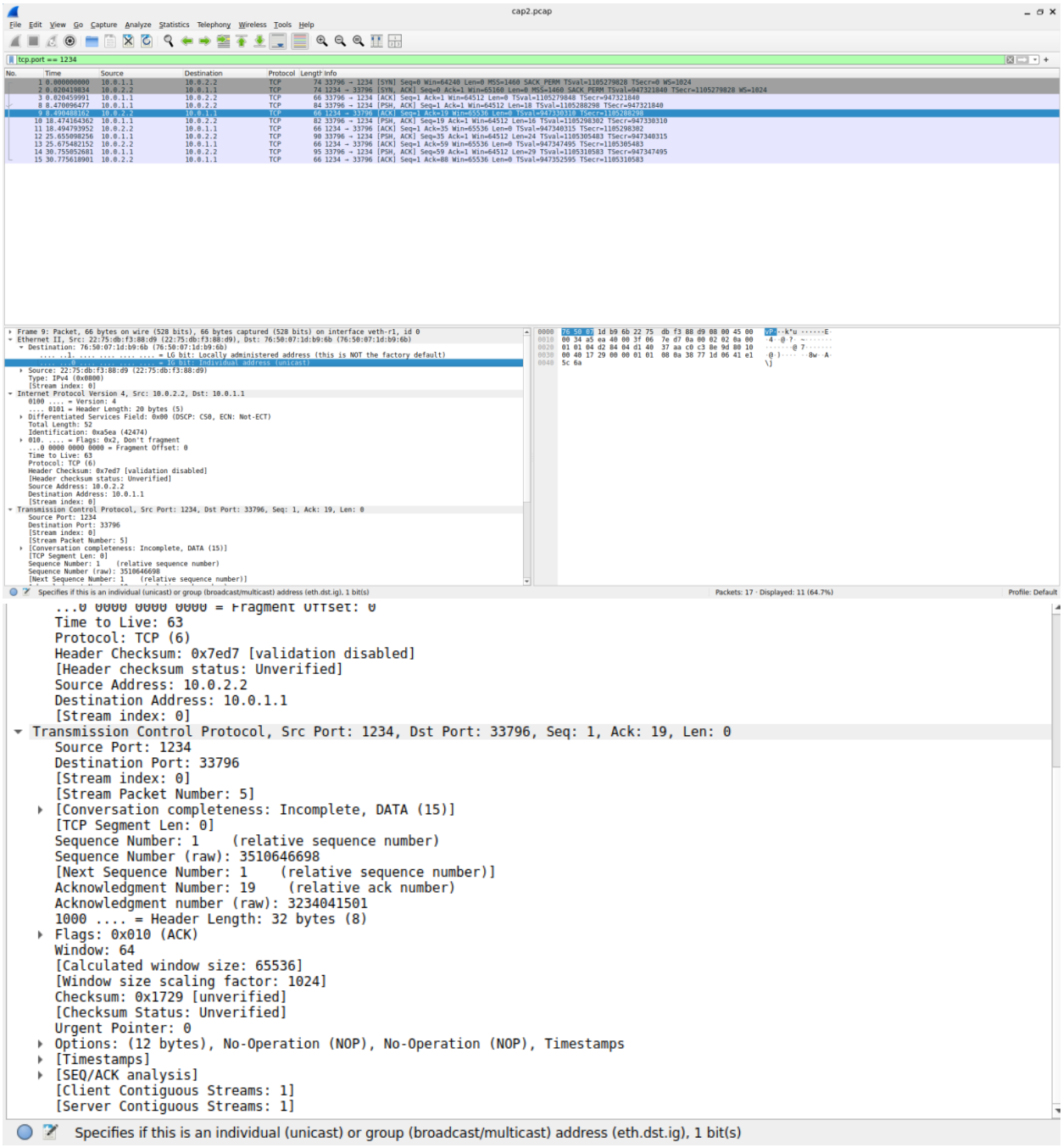

```
ttl -Y "ip"

ttl -Y "ip"

ttl -Y "ip"

1234" -T fie

6c6c6f206
20686f737
..5468697
204b6f757
0a..54687
506f696e7
707475726
00000070: 3532 3034 6336 3136 3230 610a 0a34 6436  5204c61620a..4d6
00000080: 3537 3337 3336 3136 3736 3532 3036 3736  5737361676520676
00000090: 6636 3936 6536 3732 3037 3436 3837 3236  f696e67207468726
000000a0: 6637 3536 3736 3832 3037 3236 6637 3537  f75676820726f757
000000b0: 3436 3537 3230 610a 0a                   465720a..
buntu@KOUSHAL:~$ wireshark /tmp/cap2.pcap &
[1] 3698
buntu@KOUSHAL:~$
```



Time to live :- 63

Source Address: 10.0.2.2

Destination Address: 10.0.1.1

Stage 14 – Check network delay

```
buntu@KOUSHAL:~$ wireshark /tmp/cap2.pcap &
[1] 3698
buntu@KOUSHAL:~$ tshark -r /tmp/cap1.pcap -Y "tcp.flags.syn==1 && tcp.
flags.ack==0" -T fields -e frame.time_relative 2>/dev/null
0.000000000
buntu@KOUSHAL:~$ tshark -r /tmp/cap2.pcap -Y "tcp.flags.syn==1 && tcp.
flags.ack==0" -T fields -e frame.time_relative 2>/dev/null
0.000000000
buntu@KOUSHAL:~$ tshark -r /tmp/cap3.pcap -Y "tcp.flags.syn==1 && tcp.
flags.ack==0" -T fields -e frame.time_relative 2>/dev/null
271.362435238
buntu@KOUSHAL:~$ tshark -r /tmp/cap1.pcap -Y "tcp.flags.syn==1 && tcp.
flags.ack==0" -T fields -e frame.time_epoch
1794460375.802783410
buntu@KOUSHAL:~$ tshark -r /tmp/cap2.pcap -Y "tcp.flags.syn==1 && tcp.
flags.ack==0" -T fields -e frame.time_epoch
1794460375.802800735
buntu@KOUSHAL:~$ tshark -r /tmp/cap3.pcap -Y "tcp.flags.syn==1 && tcp.
flags.ack==0" -T fields -e frame.time_epoch
1794460375.822938768
buntu@KOUSHAL:~$
```

```
buntu@KOUSHAL:~$ tshark -r /tmp/cap1.pcap -Y "tcp.analysis.retransmiss
ion" 2>/dev/null | wc -l
0
buntu@KOUSHAL:~$ tshark -r /tmp/cap2.pcap -Y "tcp.analysis.retransmiss
ion" 2>/dev/null | wc -l
0
buntu@KOUSHAL:~$ tshark -r /tmp/cap3.pcap -Y "tcp.analysis.retransmiss
ion" 2>/dev/null | wc -l
0
buntu@KOUSHAL:~$
```

Stage 15 – Check RTT

```
buntu@KOUSHAL:~$ sudo ip netns exec host1 mtr --report 10.0.2.2
Start: 2026-11-12T05:31:53+0000
HOST: KOUSHAL                     Loss%   Snt   Last   Avg  Best  Wrst
 StDev
  1.|-- 10.0.1.2                   0.0%    10    0.1   0.2   0.1   0.8
   0.2
  2.|-- 10.0.2.2                   0.0%    10   20.3  20.5  20.2  21.0
   0.2
buntu@KOUSHAL:~$
```

Stage 16 – Clean up the network

